Draft Request for Proposal – Sounder for Microwave-Based Applications

80GSFC25R7005

Amendment 23 – Mission Suitability Sections L and M Provisions Update

The following updates apply to the Mission Suitability Sections L and M provisions for the Sounder for Microwave-Based Applications procurement. In the event of any discrepancy, the final Request for Proposal provisions and contract terms will take precedence.

GSFC 52.215-210 MISSION SUITABILITY PROPOSAL INSTRUCTIONS (COMPETITIVE). (MAY 2022)

(a) The offeror's response to the requirements of this factor shall be submitted as a separate proposal volume.

(b) The volume shall be specific, detailed, and complete to clearly and fully demonstrate the offeror's understanding of the Mission Suitability subfactor requirements delineated in paragraph (d) below and the offeror's approach to effectively and efficiently accomplish those requirements, including full explanations of the techniques and procedures to be employed and the resources necessary to perform as proposed. Stating the offeror understands and will comply with the requirements, or paraphrasing the requirements is not acceptable. In addition, statements such as "standard procedures will be employed," or "well-known techniques will be used," are not acceptable. Information may not be incorporated by reference.

Enhancements:

Although the Government does not encourage/discourage enhancements to the contract's technical performance documents (e.g. Statement of Work, Specification, etc.), Offerors may choose to propose performance enhancements. In order for the Government to consider a proposed enhancement's value, the Offeror must clearly define the enhancement(s) in Contract Attachment L, "Contractor Proposed Enhancements," in performance work statement (PWS) language in accordance with the PWS instructions in Attachment L. In addition, the Offeror must describe the associated benefit(s) of the proposed enhancement(s) in its Mission Suitability Volume under the applicable Mission Suitability subfactor(s). The Offeror shall include Contract Attachment L as part of the model contract in the Contract Volume of the proposal. The Offeror may receive credit for the proposed enhancement(s) only to the extent of its description in Attachment L, and the associated benefits explained in its Mission Suitability Volume. Inconsistent statements about any enhancement(s) in an Offeror's proposal may result in a neutral or negative evaluation by the Government. Any enhancement(s) may result in a positive, neutral, or negative evaluation in spite of the Government's right to waive an enhancement(s) during contract performance in accordance with the GSFC 52.211-100, "Contractor Proposed Enhancements," clause in Section H of the contract. If the successful offer does not include any proposed enhancements, GSFC clause 52.211-100 and Attachment L will be removed from the resultant contract.

(c) The offeror shall provide, at the beginning of this volume using the table provided below, a detailed compliance matrix that cross-references the offeror's numbering structure and corresponding volume page number with all requirements associated with each subfactor delineated in paragraph (d) below. The offeror may utilize its own unique numbering structure provided the compliance matrix provides clear traceability to each of the subfactor requirements described below.

Solicitation **Mission Suitability (MS)** **Volume Requirement**	**Solicitation** **Section L Numbering Nomenclature**	**Offeror MS Volume** **Numbering Nomenclature**	**Offeror MS Volume Page Number**
Subfactor A Technical Approach	**GSFC 52.215-210, A**	Volume II	
Technical Approach Overview	GSFC 52.215-210, A1		
System Performance	GSFC 52.215-210, A2		
Instrument Design	GSFC 52.215-210, A3		
Manufacturing, Integration and Test Approach	GSFC 52.215-210, A4		
Safety and Mission Assurance	GSFC 52.215-210, A5		
Subfactor B Management Approach	**GSFC 52.215-210, B**	Volume II	
Management Approach Overview	GSFC 52.215-210, B1		
Risk Management	GSFC 52.215-210, B2		
Schedule	GSFC 52.215-210, B3		

Subcontracting	GSFC 52.215-210, B4		
Facilities	GSFC 52.215-210, B5		
Subfactor C Small Business Utilization (SBU)	**GSFC 52.215-210, C**	Volume II	

(d) The offeror shall provide a detailed response to each of the following subfactor requirements utilizing the structure and order as provided below:

Subfactor A – Technical Approach

Subfactor B – Management Approach

Subfactor C – Small Business Utilization (SBU)

The paragraph numbering, formatting, and sub-paragraphs within the subfactors below should not be construed as any indication of priority, weighting or as any establishment of elements or lower level criteria. The paragraph numbering is only provided for clarity, traceability, and ease of reading between Mission Suitability Sections L and M.

Subfactor A – Technical Approach

A.1 Technical Approach Overview

The Offeror shall provide an overview of its integrated SMBA instrument design solution with emphasis on the technical approach to meeting the Performance Requirements Document (PRD) requirements including an instrument functional block diagram.

The Offeror shall provide details of the instrument architecture, the instrument's overall layout, and overview of each instrument subsystem.

The Offeror shall describe the instrument systems engineering lifecycle and processes including engineering roles and responsibilities, technical decision-making and direction, preparation and conduction of engineering peer reviews and trade studies.

The Offeror shall describe plans for instrument internal, instrument to spacecraft interface, and ground support equipment flexibility, scalability, extensibility, modularity, accessibility, replaceability, and maintainability. The Offeror shall describe how the instrument will be minimally impacted if SMBA's environment evolves, minimally impacted during anomaly resolution, easily serviceable, and compatible with multiple spacecraft.

The Offeror shall describe the advanced technology needed to meet the SMBA requirements.

The Offeror shall describe the SMBA Instrument Technology Readiness Level (TRL) (as per NPR 7123.1D) and heritage, including all subsystems and any higher levels of assembly and substantiate the claims. The Offeror shall identify items with maturity below TRL6 and describe plans for the maturation of all technologies to TRL6. The Offeror shall describe reliance on non-SMBA programs for TRL6 and flight qualification of SMBA items. The Offeror shall describe how risks associated with these technologies will be mitigated. The Offeror shall describe its Engineering Development Program its plan for building, assembling, and testing engineering models and its plan for maintaining these engineering models through contract completion. The Offeror shall describe differences between these engineering models and the flight models. The Offeror shall also describe how the Engineering Development Program will achieve TRL6 for the SMBA assemblies and subsystems, how it will inform the design, how it will inform the manufacturing process, and how engineering models will be used after FM1 delivery.

The Offeror shall describe the approach in addressing mission life requirements.

The Offeror shall delineate Recurring Engineering (RE) and Non-Recurring Engineering (NRE) efforts and address obsolescence issues.

The Offeror shall propose a Master Equipment List (MEL) in Excel format summarizing all major subsystem components to support verification of proposed mass estimates, average power estimates, peak power estimates, envelope dimensions, data rate, data volume per orbit and per day, design heritage, and cost. The Offeror shall include key component power and thermal contribution.

A.2 System Performance

The Offeror shall provide an assessment of its proposed end-to-end SMBA instrument's compliance to the requirements (including performance and interface requirements) listed in the Performance Requirements Document (PRD) (i.e., PRD Requirements Compliance Matrix). The assessment shall include detailed system budgets and analysis with margins/contingency identified separately for all design elements spanning the beginning of life through the end-of-life timeframes.

The Offeror shall present a SMBA Frequency Plan in Excel format that provides performance metrics and resource allocations for each band. The Offeror shall describe how bands were defined subject to downlink throughput and on-board hardware constraints while maximizing science benefits and Radio Frequency Interference (RFI) detection capability. The Offeror shall include the following performance metrics: frequency and gain stability, temperature sensitivity (Noise-Equivalent Delta Temperature) including any needed instantaneous field of view aggregation, 1/f noise or striping, RFI detection algorithms, spectral resolution (for hyperspectral or RFI bands), temporal resolution, and calibration accuracy. The Offeror shall describe how continuity and new hyperspectral channels are formed, and how reprogrammability is achieved. The Offeror shall address in detail how the SMBA design implements RFI detection, hyperspectral capability, and intercalibration as well as incorporates novel technologies. The Offeror shall include a concept of operations including for its RFI detection capability.

The Offeror shall discuss the spatial sampling performance including any spatial resampling/oversampling, resolution, and coverage.

The Offeror shall discuss instrument ground and on-orbit calibration methodology and characterization approach to meet requirements pre-launch and over the mission lifetime. The Offeror shall describe any on-board calibration hardware and associated methodology for all channels including concept of operations especially any necessary on-orbit maneuvers, recommended frequency of calibration activities during check-out, normal operations and anomaly resolution; calibration during repeatability tests of the on-board calibration and the determination of the long term repeatability and stability; calibration implementation and its effect on the accuracy and short term and long term stability. The Offeror shall discuss their proposed Calibration Ground Support Equipment needed for all instrument models to include all hardware, software, and calibration targets for ambient and thermal vacuum testing. The Offeror may assume that the Government will make an external blackbody calibration target available for thermal vacuum testing.

The Offeror shall provide a radiometric error budget showing allocation to instrument requirements, including traceability to calibration uncertainty and instrument radiometric stability requirements.

A.3 Instrument Design

The Offeror shall describe the instrument elements including size, weight, power, antenna design, any scanning methodology, any detectors (analog or digital) and associated electronics, instrument electronics and data handling, thermal design, data system, and onboard calibration systems. The Offeror shall describe how redundancy, if any, has been incorporated into the instrument design and the rationale for the redundancy, or lack thereof. The Offeror shall describe any wear phenomena, potentially life-limited mechanisms, or other aspects of the design that may cause degradation of the instrument suite performance/reliability over the mission lifetime. The Offeror shall describe life tests that will be necessary for qualification of life-limited items, including how the operational and environmental conditions of the mechanism(s) will be simulated and how the timeline for the life testing fits with the overall hardware development timeline. The Offeror shall provide an assessment of the antenna pattern, sidelobes, emissivity/reflectivity. The Offeror shall describe the risks associated with developing any Field Programmable Gate Arrays (FPGAs) and/or Application-Specific Integrated Circuits (ASICs), including coding and synthesis guidelines for FPGA development especially as it relates to RFI detection, hyperspectral, and intercalibration capabilities.

The Offeror shall describe the organization of microwave sensor data coming out of the instrument, including but not limited to, data organization by band. The Offeror shall describe data formats and any pre-compression data processing to optimize the data for compression, provide the data rate analysis, and show the final uncompressed data rate calculations. The Offeror shall describe the design for any data compression and verification including associated header and error-detecting codes.

The Offeror shall describe the instrument to spacecraft interfaces including: aperture(s) and radiator fields of view (FOV), glint keep out, volume (static and dynamic), mounting interface, alignment, fault detection/notification/safing interface, pointing budgets/allocations and required spacecraft allocations, calibration requirements for the spacecraft, and contamination control requirements. The Offeror shall describe its plan on how to comply with Instrument Interface requirements, participate in the inspection of interface documents, and flag potential non-compliances, and identify and mitigate risks.

A.4 Manufacturing, Integration, and Test Approach

The Offeror shall describe unit, subsystem, and system level assembly flow in a flow chart, identifying the order of the assembly and integration. The Offeror shall describe necessary ground support equipment. The Offeror shall discuss any testing or inspections performed during the assembly and integration process. The Offeror shall describe the movement of the instrument and major subsystems among facilities as well as the preparation/packaging that is planned to assure the safety of the instrument and reduce the transportation risk to the hardware. The Offeror shall discuss how life testing of the scan drive, or any other components that require life testing, will fit within the master schedule. The Offeror shall describe how the hyperspectral radiometric testing will be performed including measurement of interchannel noise correlation. The Offeror shall describe how RFI detection testing will be performed. The Offeror shall describe in detail plans for the design, build, test, requirement verification, and delivery of the first instrument and required copies of the instrument. The Offeror shall describe how lessons learned on each build are incorporated into subsequent builds. The Offeror shall identify what activities will be accomplished in parallel and what activities need to be accomplished in close succession. The Offeror shall describe how consistency in performance is maintained for the flight model builds and how lessons learned on each build are incorporated into subsequent builds.

The Offeror shall describe the proposed plan for performance verification during development, Integration and Test (I&T) (including environmental testing), end-to-end performance testing for mission simulations and post launch test (PLT).

The Offeror shall describe the SMBA Instrument and key subsystem manufacturing approach.

The Offeror shall identify the initial long-lead and advanced procurement materials, components, and/or subsystems that could be initiated upon authorization to proceed, pending Government approval. The Offeror shall identify the rationale behind those purchases.

The Offeror shall discuss the flight qualification and the acceptance test program to be performed at the subsystem and system level, the approach for system, interface, and environmental requirements verification for each SMBA copy. Spacecraft Integration and Test and Mission Operations Support

The Offeror shall describe the necessary activities and support plans to checkout, integrate, and test the instrument with the spacecraft at the observatory level, including any special tests and equipment needed to support ambient and/or environmental testing.

The Offeror shall describe plans to meet PRD section 5.3 requirements including how and where the instruments will be stored and how the environmental conditions will be monitored. The Offeror shall discuss how these plans impact PRD section 4 requirements.

The Offeror shall describe its approach to the Instrument Simulator(s) and how they will be designed, built, integrated, and tested. The Offeror shall describe how the simulators will be maintained through the mission and how they can be used to support End-to-End testing, operational products development, mission rehearsals, and mission operations.

The Offeror shall describe how the Mission Operations Team will be supported in the development of operational products (scripts, standard and contingency operations procedures), and how they will be trained.

The Offeror shall detail the command and telemetry database development and validation process, instrument concept of operation development and list operational tools that may be needed to monitor on-orbit operations. The Offeror shall include specific information on the portions of the command and telemetry database used for RFI detection, hyperspectral, and intercalibration capabilities.

The Offeror shall identify what launch preparations and operations are needed to ensure the instrument suites are ready for launch.

The Offeror shall describe launch and early orbit operations for instrument checkout and on-orbit PLT, calibration, and check out.

A.5 Safety and Mission Assurance

The Offeror shall describe its approach for meeting the mission assurance requirements defined in Attachment A, Statement of Work, Section 3, Mission Assurance. The Offeror shall identify its approach to ensuring safe, reliable, quality hardware and software designs as well as safe, reliable, quality hardware and software builds throughout the duration of the contract. Specifically, the Offeror shall describe how it will continually monitor internal and significant subcontractors (based on the definition of significant subcontractor in the cost volume instructions) processes. The Offeror shall describe its proven mission assurance processes and practices to reduce risk. The Offeror shall provide evidence that these processes and practices result in a reliable instrument product.

Subfactor B – Management Approach

B.1 Management Approach Overview

The Offeror shall describe their overall management approach tailored to the SMBA SOW requirements and the Government’s reliance on vendor best practices in place of Government oversight and best practices. The Offeror shall discuss the organizational structure, staffing, interrelationships of technical management, business management, and subcontract and subcontractor management.

The Offeror shall describe plans for facilitating communication with Government personnel. The Offeror shall describe its vision for the balance between Government informal insight and formal oversight of the Offeror's design, development, production, integration, test, post-delivery support, and post launch support.

The Offeror shall provide a complete staffing approach that describes how the Offeror intends to staff this effort and how the approach will allow the Offeror to meet the requirements of this contract. The staffing approach shall include a comprehensive hiring plan and the risks associated with the staffing approach. The Offeror shall identify the proposed labor categories and hours required to accomplish the entire SOW requirements. The staffing approach shall include a discussion of how the staffing is adequate to conduct the SMBA flight model builds and how design expertise will be retained over the contract duration. The staffing approach shall also address any shared resources.

The Offeror shall describe any corporate resources including number of and size, as appropriate (e.g., facilities (other than those described in the technical section), equipment, financial) it intends to utilize in performance of this contract, including the availability of the resource(s) for contract-related work.

B.2 Risk Management

The Offeror shall describe its risk management process and evidence of its success. The Offeror shall list the significant expected risks along with the title, risk statement, probability, impact and severity, and acceptance or mitigation plans for each risk. The Offeror shall indicate the projected event or milestone that will close the risk. All risks discussed in other sections shall also be listed in this section.

The Offeror shall discuss how the spares program, as costed in the proposal, will mitigate risk. Provide an itemized list of spares.

B.3 Schedule

The Offeror shall provide in Microsoft Project a time-phased activity and milestone Integrated Master Schedule (IMS) referenced to requirements in the SOW detailing critical path(s), major reviews, key decision dates, procurement of long lead items and specialized components, and subsystem and instrument design, development, fabrication, assembly, integration, alignment, test, shipment, and post-delivery support. The Offeror shall describe how resources are managed and tracked including staffing, facilities utilization, and SMBA-unique ground support equipment (GSE) over the contract. The Offeror shall describe the basis of estimate for this IMS including assumptions made, any actuals from past performance, and the associated funding profile. Note that only the IMS itself is exempt from the page count, the resources management description and the basis of estimate shall be included in the portion of the mission suitability volume that is included in the page count limitation.

B.4 Subcontracting

The Offeror shall describe its strategy for using (or not using) significant subcontractors (based on the definition of significant subcontractor in the cost volume instructions). If significant subcontractors are proposed, identify their interfaces to your organizational structure and provide: 1) the nature and extent of the work to be performed by the significant subcontractor, including split of responsibilities and the potential percentages of work to be performed, and 2) methods of management and reporting to GSFC of subcontractors' financial and technical plans and performance. Indicate alternate sources for major and critical subsystems/components that are purchased or that are provided by subcontractors or other team members. The Offeror shall discuss its plans for addressing any problems (technical and programmatic) that arise as a result of the proposed subcontract strategy and organization structure or poor and/or non-performance of subcontracted portions of the contract. The Offeror shall discuss plans for supporting Government insight into subcontracted portions of the effort.

B.5 Facilities

The Offeror shall discuss its proposed facilities to accommodate design, development, manufacturing, integration, testing, delivery, and storage of all instruments (including engineering models and all instruments). The Offeror shall describe which (if any) antenna ranges, Electro Magnetic Interference (EMI) test facilities, vibration test facilities, thermal vacuum chambers, clean rooms, and other development, manufacturing, and test environments it will use. Included in this discussion, the Offeror shall compare the range of capabilities of these facilities to the full range needed to test and calibrate the Instrument. The Offeror shall describe the size, number, and capabilities of the identified facilities and how the facilities meet the needs for parallel activities as defined in the IMS. The Offeror shall discuss where these facilities are located, the facilities' capacity for accommodating SMBA instruments including SMBA instruments' specific environmental storage requirements, when these facilities will be necessary, their current state, and how the Offeror will ensure their availability.

Facility readiness and capability for integration and test of Instrument in all areas above shall include suitability of facilities for instrument size, instrument complexity, verifying instrument requirements, and instrument safety, including safe transport.

Subfactor C – Small Business Utilization (SBU)

All Offerors, except small businesses, shall complete the portion of the instructions under Small Business Subcontracting Plan. Small businesses are not required to submit Small Business Subcontracting Plans; however, small businesses shall indicate the amount of effort proposed to be done by a small business either at the prime level or at the first-tier subcontract level.

All Offerors (small businesses and other than small businesses) shall respond to the Commitment to Small Business Program.

(A) Small Business Subcontracting Plan

(1) This solicitation contains FAR clause 52.219-9 ALTERNATE II, "Small Business Subcontracting Plan". The Plan described and required by the clause, including the associated subcontracting percentage goals and subcontracting dollars, shall be submitted with the proposal.

(2) The Contracting Officer's assessment of appropriate subcontracting goals for this acquisition, expressed as a percent of the TOTAL CONTRACT VALUE, is as follows:

Small Businesses (SB)	8.7%
Small Disadvantaged Business (SDB) Concerns	0.8%
Women-Owned Small Business (WOSB) Concerns	1.0%
Historically Black Colleges and Universities (HBCU)/ Minority Serving Institutions (MSI)	0.05%
HUBZone Small Business (HBZ) Concerns	0.1%
Veteran-Owned Small Business (VOSB) Concerns	0.6%
Service-Disabled Veteran-Owned Small Business (SDVOSB) Concerns	0.1%

(3) The numbers above reflect the Contracting Officer's assessment of the appropriate subcontracting goals to be achieved at the completion of contract performance. When appropriate, an Offeror may discuss plans to phase-in small business concerns, explaining the rationale for the phase-in schedule. If the Offeror anticipates that it will not meet the proposed small business goals by the time the Offeror submits its first Individual Subcontracting Report (ISR) for this effort as required by FAR clause 52.219-9 ALTERNATE II, Small Business Subcontracting Plan, the Offeror should discuss its approach to include a timeline for meeting these goals and the associated rationale.

(4) Offerors are encouraged to propose goals that are equivalent to or greater than those that the Contracting Officer has recommended. However, Offerors shall perform their own independent assessments of the small business subcontracting opportunities and are encouraged to propose goals exceeding the recommended goals where practical.

(5) The Plan that the Offeror submits with its proposal shall be incorporated in Section J as Attachment G in the resulting contract. The requirements in the Plan shall flow down to first-tier large business subcontracts expected to exceed $900,000 (or $2,000,000 for construction of a public facility). Although these first-tier large business subcontractors are encouraged to meet or exceed the stated goals, NASA recognizes that the subcontracting opportunities available to these subcontractors may differ from the recommended goals in the solicitation based upon the nature of their respective performance requirements.

(6) Offerors are advised that a proposal will not be rejected solely because the Plan submitted does not meet the NASA recommended goals listed in paragraph (a) (2) above in terms of percent of the TOTAL CONTRACT VALUE. NASA will consider the amount of work that the

prime contractor plans to perform when determining whether a subcontracting plan is acceptable. Offerors shall discuss the rationale for any goal proposed that is less than the Contracting Officer's recommended goal in any category. In addition, the Offeror shall describe the efforts made to establish a goal for that category and what ongoing efforts, if any, the Offeror plans during contract performance to increase participation in that category.

(7) In addition to submitting a Small Business Subcontracting Plan in accordance with the Section I, FAR clause 52.219-9 Alternate II, Offerors shall complete **Exhibit 8**, SMALL BUSINESS SUBCONTRACTING PLAN GOALS, which provide a breakdown of the Offeror's proposed goals, by small business category, expressed in terms of both a percent of the TOTAL CONTRACT VALUE and a percent of TOTAL PLANNED SUBCONTRACTS. TOTAL CONTRACT VALUE includes CLIN 0001, CLIN 0002, and CLIN 0003 as defined in contract clause GSFC 52.211-91 SCOPE OF WORK.

(NOTE: FOR PURPOSES OF THE SMALL BUSINESS SUBCONTRACTING PLAN, THE PROPOSED GOALS SHALL BE STATED AS A **PERCENT OF TOTAL SUBCONTRACTS**, NOT AS A PERCENT OF THE TOTAL CONTRACT VALUE, REFER TO THE BELOW EXAMPLE)

Example of Subcontracting Goals as expressed in both the Total Contract Value and Planned Subcontract Value for a Total Contract value of $100M and planned subcontract value of $50M.

	Column A	***Column B***	***Column C***
Category	**Percent of Total Contract Value**	**Dollar Value**	**Percent of Subcontracting Value**
Small Business (SB) Concerns	25 percent	$25,000,000	50 percent
Other than Small Business Concerns	N/A	$25,000,000	50 percent
Total Dollars to be Subcontracted	N/A	$50,000,000	100 percent
The following small business subcategories do not necessarily add up to the percentage and dollar amount in the "Small Business Concerns" category above, since some small businesses do not fall into any of the subcategories below, while others will fall into more than one subcategory below.			
Small Disadvantaged Business (SBD) Concerns	5.5 percent	$5,500,000	11 percent
Women-Owned Small Business (WOSB) Concerns	9 percent	$9,000,000	18 percent

Historically Black Colleges and Universities (HBCU)/Minority Serving Institutions (MSI)	1.5 percent	$1,500,000	3 percent
HUBZone (HBZ) Small Business Concerns	1.5 percent	$1,500,000	3 percent
Veteran-Owned Small Business (VOSB) Concerns	2.5 percent	$2,500,000	5 percent
Service-Disabled Veteran-Owned Small Business (SDVOSB) Concerns	1.5 percent	$1,500,000	3 percent

The Offeror proposes small business subcontracting goals as a percentage of the Total Contract Value in column A.

Then based on the $100 million Total Contract Value, the resulting statement of dollars that the Offeror would include in the Subcontracting Plan, as required by paragraph (d)(2) of FAR clause 52.219-9 Alternate II, would be as indicated in column B.

However, the Small Business Subcontracting Plan shall also express goals as a percent of total planned subcontracts. Assuming total subcontracting of $50M, the resulting percentage goals, expressed as a percent of total subcontract dollars would be stated in the Small Business Subcontracting Plan as required by paragraph (d)(1) of FAR clause 52.219-9 ALTERNATE II in column C.

(B) Commitment to the Small Business Program

(1) All Offerors must briefly describe the work that small businesses will perform to meet the stated requirements. Proposals shall also identify any work to be subcontracted that is considered "High-Tech". "High-Tech" is defined as research and development efforts that are within or advance the state-of-the-art in technology discipline and are performed primarily by professional engineers, scientists, and highly skilled and trained technicians or specialists.

(2) If the subcontractor(s) is known, Offerors shall describe the work that each subcontractor will perform and specify the extent of commitment to use the subcontractor(s) (enforceable vs. non-enforceable commitments). (Small Business Offerors shall provide this information to the extent that subcontracting opportunities exist in their approach to performing the requirement.)

(3) All Offerors shall provide information demonstrating the extent of commitment to utilize small business concerns and to support their development. Information provided shall include a brief description of established or planned procedures and organizational structure for Small Business outreach, assistance, counseling, market research and Small Business identification, and relevant purchasing procedures. (For Other than Small Business Offerors, this information

shall conform to applicable portions of the submitted Small Business Subcontracting Plan. Small Business Offerors shall provide this information to the extent subcontracting opportunities exist in their approach to performing the requirement.)

(4) The NASA Mentor-Protégé Program is designed to incentivize NASA large prime contractors to assist a Small Disadvantaged Business, a Women-Owned Small Business, a HUBZone Small Business, a Veteran-Owned or Service-Disabled Veteran-Owned Small Business, or Historically Black College and University/Minority Serving Institution in enhancing their capabilities to perform NASA contracts and subcontracts, foster the establishment of long-term business relationships between these entities and NASA large prime contractors, and increase the overall number of these entities that receive NASA contract and subcontract awards. Provide a description of the prime's planned participation in the NASA Mentor Protégé Program.

(e) Offerors shall ensure the information submitted in this volume is consistent with information submitted in other volumes, as applicable.

(End of provision)

GSFC 52.215-311 MISSION SUITABILITY EVALUATION FACTOR. (MAY 2022)

(a) In accordance with FAR Subpart 15.2 and NFS Subpart 1815.2, the Government will evaluate the offeror's demonstrated understanding of the Mission Suitability subfactor requirements and approach for accomplishing those requirements, the appropriateness of the offeror's proposed resources, and associated programmatic risk. A lack of resource realism may adversely affect the offeror's Mission Suitability evaluation and result in cost realism adjustments under the Cost factor. The Mission Suitability evaluation will take into consideration whether the resources proposed are consistent with the proposed efforts and accomplishments associated with each subfactor or whether they are overstated or understated for the effort to be accomplished as described by the Offeror and evaluated by NASA. The Government will validate the consistency between all proposal volumes and any inconsistencies identified may indicate a lack of understanding and adversely impact the offeror's adjectival rating(s) and score. Only that information provided within the proposal will be evaluated; any reference to previously submitted information, if any, will be considered only to the extent the information is resubmitted as part of the proposal. Information incorporated by reference will not be considered or evaluated.

The Government may choose to incorporate any positive aspects of an Offeror's approach to meeting/exceeding contract requirements into the final contract, particularly if any positive proposal area results in Strength or Significant Strength findings in the Mission Suitability evaluation. An Offeror's proposed Mission Suitability approach shall be consistent with its proposed cost/price information.

If an Offeror elects to propose enhancements in accordance with the Mission Suitability Volume instructions in provision GSFC 52.215-210, the Offeror's completed Contract Attachment L, "Contractor Proposed Enhancements," and the description of the associated benefits for each proposed enhancement under the applicable Mission Suitability subfactor, will be evaluated for reasonableness, effectiveness, and overall performance benefit.

Subfactor A – Technical Approach

The Government will evaluate the Offeror's Technical Approach, including all areas and information, in Section L, GSFC 52.215-210, Subfactor A, for adequacy, clarity, completeness, compliance, effectiveness, efficiency, extensibility, flexibility, heritage, innovation, reasonableness, robustness, scalability, soundness, substantiation, validation, degree of risk, and likelihood of successful implementation.

Subfactor B – Management Approach

The Government will evaluate the Offeror's Management Approach, including all areas in Section L, GSFC 52.215-210, Subfactor B, for adequacy, appropriateness, clarity, completeness, effectiveness, efficiency, flexibility, heritage, innovation, realism, reasonableness, degree of risk, and likelihood of successful implementation.

Subfactor C – SMALL BUSINESS UTILIZATION (SBU)

NASA will evaluate the Small Business Subcontracting Plan, as required by FAR clause 52.219-9 ALTERNATE II, “Small Business Subcontracting Plan,” which applies to all Offerors, except small businesses. The evaluation of Commitment to Small Business Program applies to all Offerors.

(A) Small Business Subcontracting Plan

(1) The Small Business Subcontracting Plan will be evaluated in terms of the Offeror’s proposed subcontracting goals for first-tier subcontractors (overall subcontracting goals and individual subcontracting goals by category) in comparison to the Contracting Officer’s assessment of the appropriate subcontracting goals for this procurement. The Offeror's Small Business Subcontracting Plan will also be evaluated in terms of meeting the requirements of FAR 19.206-2 Elements of the Subcontracting Plan. NASA will consider the amount of work that the prime contractor plans to perform when determining whether a subcontracting plan is acceptable. The evaluation of the Small Business Subcontracting Plan will be based on TOTAL CONTRACT VALUE.

(2) Since Small Businesses are not required to submit subcontracting plans, NASA will only evaluate the amount of work that the small business prime and any small businesses proposed to perform at the first-tier subcontract level. The proposed amount of work that the prime small business and first-tier small business subcontractors will perform will be evaluated against the Contracting Officer’s assessment of the overall subcontracting goal for this procurement. For small business primes and their first-tier subcontractors, individual subcontracting goals by small business categories will not be evaluated.

(B) Commitment to Small Businesses

(1) NASA will evaluate the extent to which the work performed by a small business subcontractor(s) is “high-tech” work as defined in NFS 1819.001. NASA also will evaluate the extent of commitment to use the subcontractor(s) (enforceable vs. non-enforceable commitments).

(2) NASA will evaluate the extent to which the identity of the small business subcontractor is specified in the proposal as well as the extent of the commitment to use small businesses. (For Small Business Offerors, NASA will evaluate this only if subcontracting opportunities exist.)

(3) NASA will evaluate the Offeror’s established or planned procedures and organizational structure for small disadvantaged business (SDB) outreach, assistance, counseling, market research and SDB identification, and relevant purchasing procedures. (For Other than Small Business Offerors, this information should conform to its submitted Small Business Subcontracting Plan. For Small Business Offerors, NASA will evaluate this only if subcontracting opportunities exist.)

(4) NASA will evaluate the Offeror’s participation and/or proposed participation in the Mentor Protégé program and their planned commitment to enter into mentor-protégé agreements to provide appropriate developmental assistance to enhance the protégé’s ability to perform successfully under contracts and/or subcontracts.

(b) Subfactors will be numerically weighted and scored in accordance with the 1,000-point scale provided below.

Subfactor	Points
Subfactor A – Technical Approach	500
Subfactor B – Management Approach	400
Subfactor C – Small Business Utilization (SBU)	100
Total	1,000

(c) In accordance with FAR 15.001, weaknesses, significant weaknesses, and deficiencies will be assessed based on the following definitions:

Weakness means a flaw in the proposal that increases the risk of unsuccessful contract performance.

Significant weakness in the proposal is a flaw that appreciably increases the risk of unsuccessful contract performance.

Deficiency is any part of an offer that does not conform to a material requirement of a RFP. A material requirement is one that affects price, quantity, quality, or delivery, or that the RFP requires to be met at the time of proposal submission.

Additionally, strengths and significant strengths will be assessed based on the following definitions:

Strength is an aspect of the proposal that will have some positive impact on the successful performance of the contract.

Significant strength is some aspect of the proposal that greatly enhances the potential for successful contract performance.

(d) The Government will assign adjectival ratings and numerical scores for each subfactor.

(e) The Government will establish a total Mission Suitability factor numerical score by adding all of the subfactor points assessed. A summary adjectival rating will not be assigned for this factor.

(End of provision)